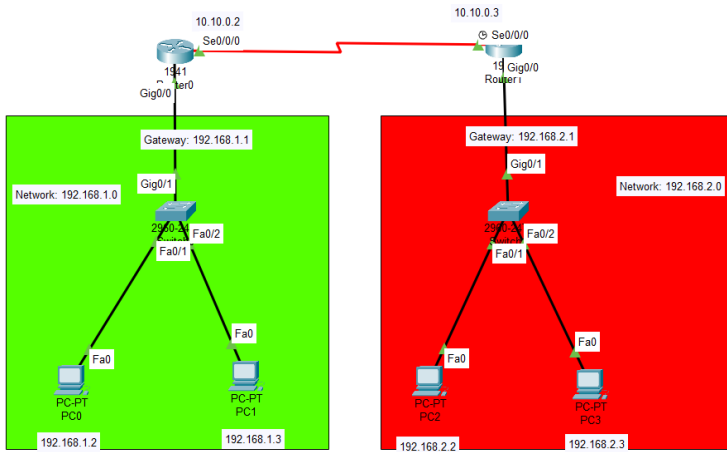


Configuration of ARP and Static Routing using Cisco network switch and verify the connectivity



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Devices and Connections

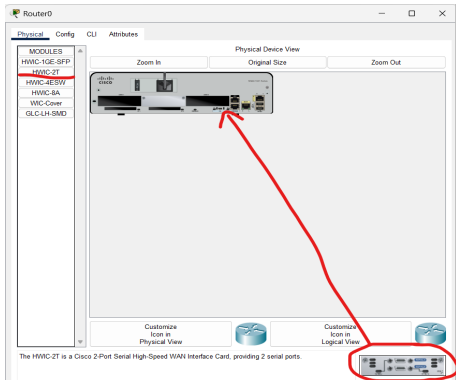
- Select two routers (Model: 1941) and place them on the workspace
- Select two switches (Model: 2960) and place them on the workspace
- Select four PCs and place them on the workspace
- Connections for left-side switch
 - Use a Copper straight-through and connect
 - FastEthernet of PC0 $\Rightarrow$ FastEthernet0/1 of the switch
 - FastEthernet of PC1 $\Rightarrow$ FastEthernet0/2 of the switch
 - GigabitEthernet0/1 of switch to GigabitEthernet0/0 of the left-side router
- Connections for right-side switch
 - Use a Copper straight-through and connect
 - FastEthernet of PC2 $\Rightarrow$ FastEthernet0/1 of the switch
 - FastEthernet of PC3 $\Rightarrow$ FastEthernet0/2 of the switch
 - GigabitEthernet0/1 of switch to GigabitEthernet0/0 of the right-side router

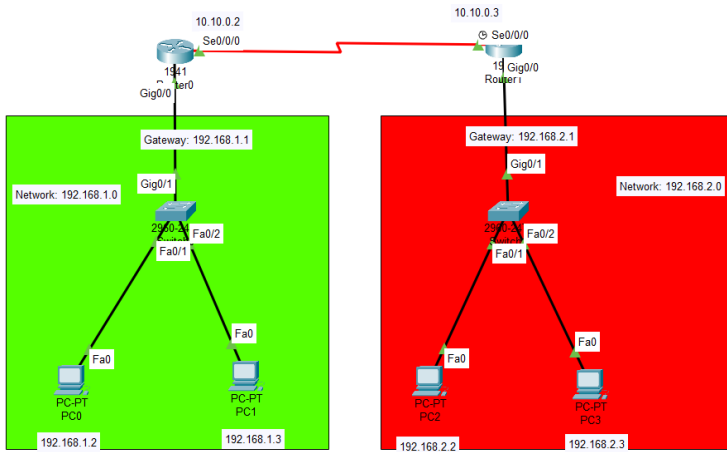


Router -Router Connection

Serial ports are required for connecting the routers

- Switch-OFF the Router
- Drag and Drop the HWIC-2T module on the empty rack of the router
- Switch-ON the Router
- Repeat this step for the second router
- Use Serial DTE wire to connect the serial ports of both the routers (Se0/0/0 ↔ Se0/0/0)







Assign IP Address

Click on PC → Desktop Tab → IP Configuration

- PC0: IPv4: 192.168.1.2; SubMask:255.255.255.0; Gateway: 192.168.1.1
- PC1: IPv4: 192.168.1.3; Sub Mask:255.255.255.0; Gateway: 192.168.1.1
- PC2: IPv4: 192.168.2.2; Sub Mask:255.255.255.0; Gateway: 192.168.2.1
- PC3: IPv4: 192.168.2.3; Sub Mask:255.255.255.0; Gateway: 192.168.2.1



Configuring Gigabit Interface of router (Left)

Router0

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0**
- GigabitEthernet0/1
- Serial0/0/0
- Serial0/0/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.2F94.D801

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```



Configuring Serial Interface of router (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- Serial0/0/0**
- Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.10.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 10.10.0.2 255.0.0.0
Router(config-if)#ip address 10.10.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
```




Configuring Static routing table of the router (left)

The screenshot shows the configuration interface for Router0. The left sidebar contains a tree view with the following categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), SWITCHING (VLAN Database), and INTERFACE (GigabitEthernet0/0, GigabitEthernet0/1, Serial0/0/0, Serial0/0/1). The 'Static' option under ROUTING is selected and highlighted with a red arrow. The main area is titled 'Static Routes' and contains a table with the following entries:

Network	Mask	Next Hop
192.168.2.0	255.255.255.0	10.10.0.3

Red arrows point to the 'Add' button and the 'Next Hop' field. Below the table, there is a section for 'Network Address' with the text '192.168.2.0/24 via 10.10.0.3'. A 'Remove' button is located at the bottom right of this section.

Equivalent IOS Commands

```
LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 10.10.0.2 255.0.0.0
Router(config-if)#ip address 10.10.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#ip route 192.168.2.0 255.255.255.0 10.10.0.3
Router(config)#
```



Configuring Gigabit Interface of router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0**
- GigabitEthernet0/1
- Serial0/0/0
- Serial0/0/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 00E0.F9B6.4201

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 192.168.2.1 255.255.255.0
Router(config-if)#
```



Configuring Serial Interface of router (Right)

Router1

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- Serial0/0/0**
- Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.10.0.3

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
ip address 192.168.2.1 255.255.255.0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
ip address 10.10.0.3 255.0.0.0
Router(config-if)#
```



Configuring Static routing table of the router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
 - Static**
 - RIP
- SWITCHING**
 - VLAN Database
- INTERFACE**
 - GigabitEthernet0/0
 - GigabitEthernet0/1
 - Serial0/0/0
 - Serial0/0/1

Static Routes

Network 192.168.1.0

Mask 255.255.255.0

Next Hop 10.10.0.2

Add

Network Address

192.168.1.0/24 via 10.10.0.2

Remove

Equivalent IOS Commands

```
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
ip address 10.10.0.3 255.0.0.0
Router(config-if)#ip address 10.10.0.3 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 10.10.0.2
Router(config)#
```



Check Connection

Ping the PCs to check connection. Every PC is connected to every other PC

For multiple networks:

https://youtu.be/rZw_b0wpQ00?si=CTX54aCtjHj8iKcE



Inspect Routing Tables

Select Inspect Icon $\Rightarrow$ Click on the router to check routing table

Cisco Packet Tracer - C:\Users\ompg6\OneDrive\Documents\NPS_24_25\Hands_On\exp7fin.pkt

File Edit Options View Tools Extensions Window Help

Logical (Physical) x: 510, y: 181

Routing Table for Router0

Type	Network	Port	Next Hop IP	Metric
C	10.0.0.0/8	Serial0/0/0	---	0/0
L	10.10.0.2/32	Serial0/0/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0
S	192.168.2.0/24	---	10.10.0.3	1/0



Inspect ARP Tables

ARP is generally automatically handled by Cisco switches and routers.
Select Inspect Icon $\Rightarrow$ Click on the router to check ARP table

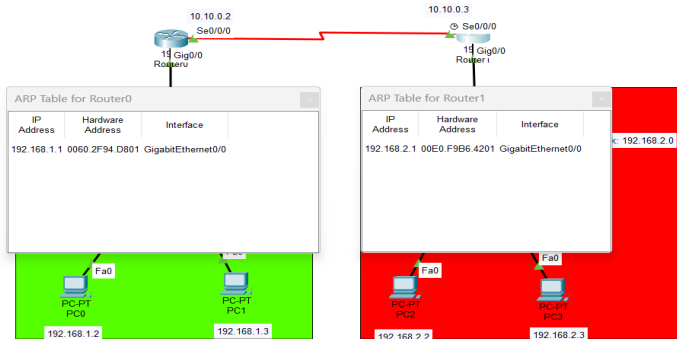


Figure: Initial Table



Inspect ARP Tables

Select Inspect Icon $\Rightarrow$ Click on the router to check ARP table

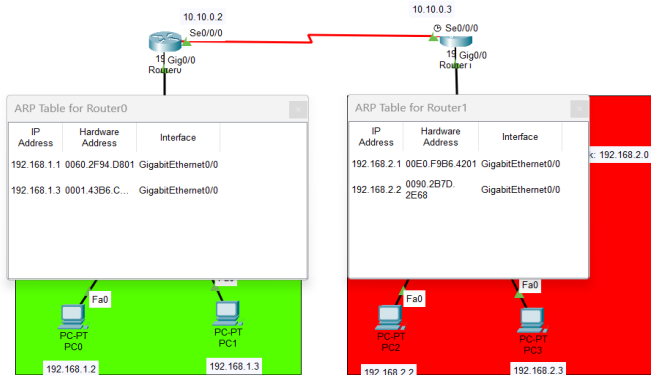


Figure: Table updated after sending message from PC1 to PC2



Inspect ARP Tables

Select Inspect Icon $\Rightarrow$ Click on the router to check ARP table

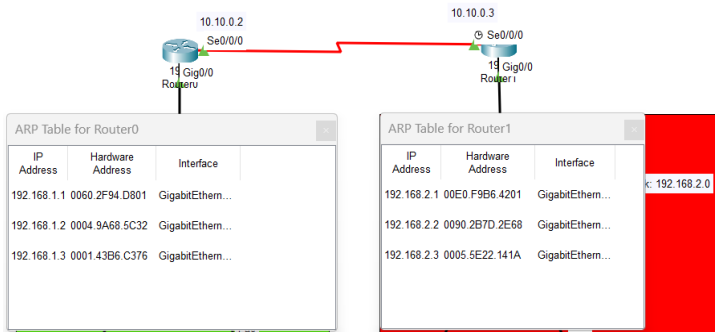


Figure: Table updated after sending message from PC0 to PC3



Conclusion

You have configured static routing for managing network traffic efficiently and ensuring proper connectivity in a multi-network environment